THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 06

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 720	$\frac{90\times144}{18}=720$
2	**(c)** $17\times19\times23$	$17\times19=323$, $323\times23=7429$
3	**(a)** x^2-4x	$x(x-4)$
4	**(b)** Student A is correct	$x^2-0x+(-5)$
5	**(d)** $(6,4)$	$2x=12$
6	**(b)** Incorrect — they intersect at $(2,2)$	One is vertical, the other horizontal
7	**(b)** −4	$16-20$
8	**(c)** −8	$4+2k+12=0$
9	**(b)** 59	$3+14\times4$
10	**(c)** $-2,-2,-2,\ldots$	Constant terms
11	**(b)** 8 cm	$\frac{3}{4}=\frac{6}{EC}$
12	**(c)** 50°	A corresponds to Q
13	**(a)** 45°	$\cot45^\circ=1$
14	**(d)** 24 cm	$\sqrt{676-100}$
15	**(a)** 60°	$\frac{22}{132}\times360^\circ$
16	**(a)** 37.68 cm²	$\frac{1}{3}\times113.04$
17	**(b)** 308 cm³	154×2
18	**(a)** 35	Mid-value
19	**(b)** 24	$\frac{3(26)-30}{2}$
20	**(c)** $\frac{5}{9}$	E, U, A, I, O

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$40=2^3\times5$, $36=2^2\times3^2$, $126=2\times3^2\times7$; HCF $=2$, LCM $=2^3\times3^2\times5\times7=2520$	1+1
22	$4+2-6=0$, $9-3-6=0$ ⇒ both are zeroes; sum $1=-\frac{-1}{1}$ ✓	1+1
23	$D=33$ ⇒ $x=\frac{-1\pm\sqrt{33}}{4}$	1+1
24	$\angle C=40^\circ$, $\angle P=60^\circ$ ⇒ $\triangle ABC\sim\triangle PQR$ (AA)	1+1
25	$9+(k-3)^2=25\Rightarrow k-3=\pm4$ ⇒ $k=7$ or -1	1+1
26	$BC=12$; $\sin A=\frac{12}{13}$, $\tan C=\frac{5}{12}$	1+1
27	$30\tan30^\circ=10\sqrt{3}\approx17.32$ m	2
28	$\angle AOB=110^\circ$; $\triangle OAB$ isosceles ⇒ $\angle OAB=35^\circ$	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Points $(0,9),(3,0)$; $(0,1),(-1,0)$. Intersect at $(2,3)$ ⇒ consistent	2+1
30	$a=35$; $\Sigma fd=-40-40+0+50+40=10$ ⇒ $35+\frac{10}{20}=35.5$	1+1+1
31	$11,13,\ldots,99$: 45 terms; $S=\frac{45}{2}(110)=2475$	1+1+1
32	$r=\sqrt{169-144}=5$ cm; nearest distance $=13-5=8$ cm	2+1
33	240°: $\frac{2}{3}\times\frac{22}{7}\times110.25=231$ cm²	1+1+1
34	$\frac{3}{26}$; $\frac{16}{52}=\frac{4}{13}$; $\frac{3}{4}$	1+1+1
35	Width 12 m; height above window $12\sqrt{3}$ ⇒ $12(1+\sqrt{3})\approx32.78$ m	1+1+1
	OR: $60\sqrt{3}+20\sqrt{3}=80\sqrt{3}\approx138.56$ m	1+1+1

36	$n = 40$; cf 6, 15, 25 ⇒ $40 + \frac{20-15}{10} \times 20 = 50$; the student is wrong	1+1+1
	OR: $x + y = 15$; $28.5 = 20 + \frac{30-(5+x)}{20} \times 10 \Rightarrow x = 8,\ y = 7$	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) 10 (ii) $(4, 5)$ (iii) $(3, 1)$ and $(5, 1)$ (iv) both distances 5 ✓	1+1+2+1
	OR: $\frac{3x+8}{7} = -1 \Rightarrow x = -5$; $\frac{3y+20}{7} = 2 \Rightarrow y = -2$ ⇒ B(−5, −2) (3); $AB = 7\sqrt{2}$ (2)	3+2
38	$\sec^2 A(\sec^2 A - 1) = (1 + \tan^2 A)\tan^2 A$ (3); $4 - 2 = 2 = 1 + 1$ ✓ (2)	3+2
	OR: $2 + 2\tan^2 A = 2\sec^2 A$ (3); $2 \times 4 = 8$ (2)	3+2
39	(i) $385 + 89.83 + 154 = 628.83$ cm³ (ii) $l = 12.5$: $220 + 77 + 137.5 = 434.5$ cm²	3+2
	OR: (i) 282.6 m³ (ii) 37.68 m³ (iii) 320.28 m³ (iv) $l = 5$: $188.4 + 47.1 = 235.5$ m²	1+1+1+2
40	(i) $a = 52.5$: $\Sigma fu = 19$ ⇒ 54.4 kg (ii) cf 5, 13 ⇒ $50 + \frac{12}{14} \times 5 \approx 54.29$ kg (iii) $50 + \frac{6}{28-8-13} \times 5 \approx 54.29$ kg	2+2+1
	OR: $a = 32.5, h = 5$; $\Sigma fu = -25$ ⇒ mean $= 32.5 - \frac{125}{35} \approx 28.93$ (3). Mode $= 30 + \frac{1}{20-9-3} \times 5 = 30.625$ (2)	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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